TIC 237657109

Eclipsing binary candidate dossier - Star Vetter

CLEAN

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BLS EPHEMERIS
  Period       2.545124 d
  Epoch (t0)   1469.5678 BTJD
  Duration     0.1700 d
  Depth        1.553 %  (15,529 ppm)
  SNR          56.3

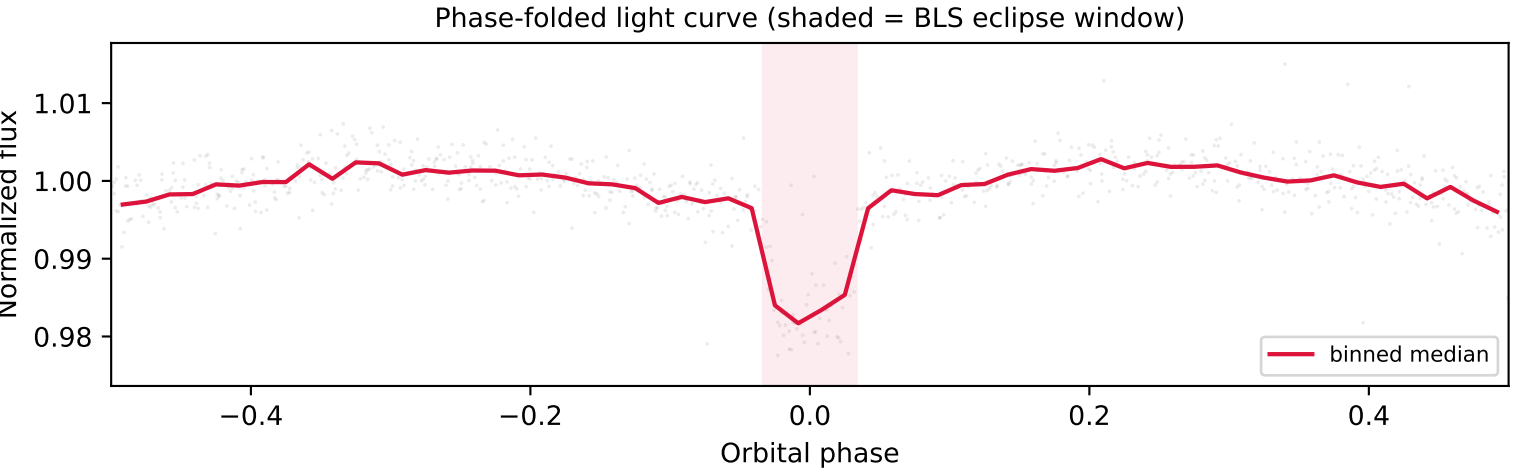
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     8  (gate >= 3)
  Robust depth ratio    1.08  (gate >= 0.50)
  Robust depth sigma    36.4  (gate >= 7.0)
  Odd/even z-score      1.72  (consistent)
  Secondary eclipse     not detected
```

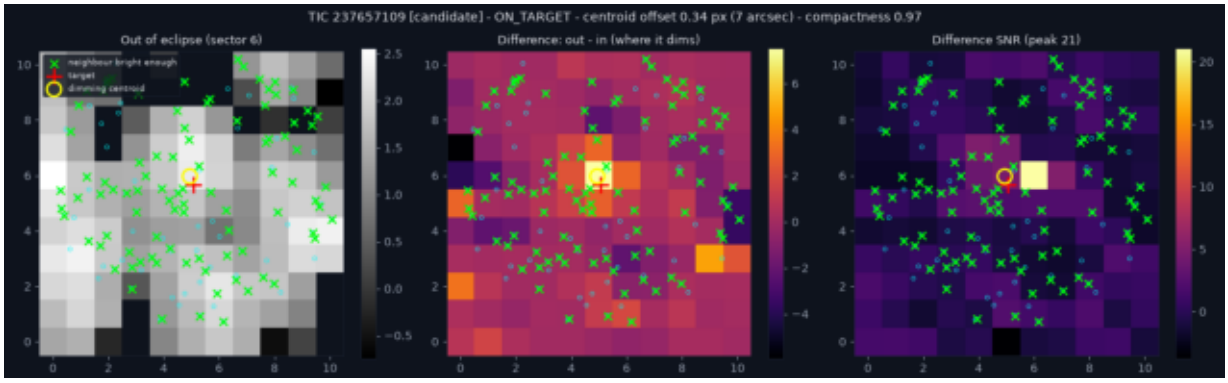
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 6    light-curve dip in this sector: 49.6 sigma    t0 refined by 0.0000 phase
  Difference-image peak SNR   21.09   (gate >= 5.0)      Compactness   0.97   (gate >= 0.50)
  Dimming centroid offset     0.34 px (7 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 237657122
    0.48 px away, dTmag -1.08 - farther from the dimming than the target
```





No caveats - clean detection on every check.